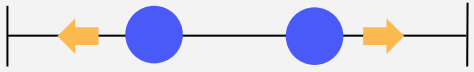
Constant Force Interacting System

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Introduction

The Problem

Two identical particles repelling each other with a constant force, trapped in a box of width a (in 1D).



$$V(x_1, x_2) = -g\left(|x_2 - x_1| - a\right)$$

This system describes two charges where both the particles and the electric field are confined to one dimension.

The Goal

Our goal was to model the behaviour of the system as the strength, *g*, of the interaction force changes.

Analytic Solution in Low and High *g* Limits

Low *g*

The $g = 0$ case reduces to the well known infinite square well, which has energies of the form

$$E = \frac{\hbar^2\pi^2(n_1^2 + n_2^2)}{2ma^2}.$$

At low *g*, corrections to the infinite square well energies are given by time-independent perturbation theory. They are rational functions in n_1 and n_2.

High *g*

In the case where *g* is very large, the Hamiltonian becomes

$$\hat{H} \to -\frac{\hbar}{2m}\left(\frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2}\right) + g(x_1 + x_2),$$

the solutions to which are products of Airy functions. These group into superpositions of two or four separable terms to maintain the symmetries of the original equation. They have energies

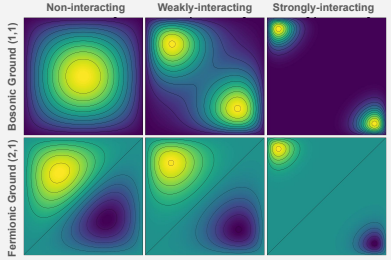
$$E = \sqrt[3]{\frac{\hbar^2 g^2}{2m}}(z_1 + z_2) \quad \text{for} \quad z_1, z_2 \in Z,$$

where Z is the set of zeroes of the Airy functions.

Numerical Solutions for all Cases

Method

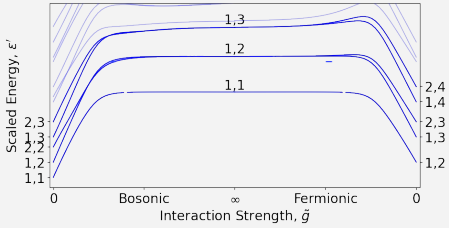
Using the finite-difference approximation, the Schrödinger equation was discretized and solved as an eigenvalue problem.



Spectrum

The degeneracy in the energies increases with *g*.

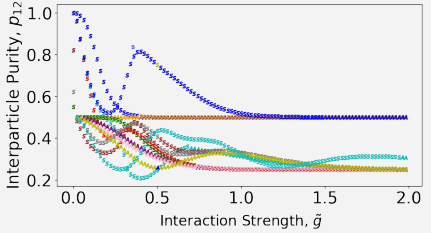
This exhibits a phenomenon known as fermionization.



Purity

The purity has the anticipated values at low and high *g*, but not in between.

The fermionic ground state's purity is constant, and most do not have monotonic purity.



Interparticle Purity in Low and High *g* Limits

Definition

Interparticle purity measures the entanglement (correlation) of two particles. It is inversely related to entanglement; a purity of 1 corresponds to completely uncorrelated particles.

Calculation

The purity is given by the trace of the squared reduced density matrix, which is the sum of the Schmidt coefficients to the fourth.

Thus, the purity is 1 or 1/2 at $g = 0$, and 1/2 or 1/4 at high *g* — depending on whether the wave function is a superposition of one, two, or four states.

Conclusion

We have analytically explained the behaviour of the system in the low- and high-interaction limits. We have obtained computational results which agree at the limits and bridge the gap between them.

Further work includes developing a more quantitative explanation of the entanglement in the non-limiting cases.

Acknowledgements

This work was supported by the Delord-Mockett Student Research Award.